

Global Air Pollution Analysis

Shubhanshu Dubey, Sneha Chepurwar, Divyanjali Gopishetty

Mining Minds

email@example.com

Abstract—This study presents a comprehensive analysis of global air pollution patterns using a dataset of 23,463 cities across 175 countries. We analyzed four major pollutants: particulate matter (PM2.5), carbon monoxide (CO), ozone (O₃), and nitrogen dioxide (NO₂). Our analysis reveals that 18.4% of monitored cities have concerning air quality levels (unhealthy or worse), with PM2.5 emerging as the dominant pollutant (mean AQI: 68.52). Geographic analysis identified South Asia, particularly India and Pakistan, as the most severely affected region, with multiple cities reaching the maximum measurable AQI of 500. Correlation analysis revealed moderate positive relationships between pollutants, with CO and NO₂ showing the strongest correlation ($r = 0.488$), suggesting common emission sources. Ozone demonstrated distinct geographic patterns due to its photochemical formation mechanism. Our findings indicate that 2,556 cities (10.9%) have concerning PM2.5 levels, affecting millions of people. The study provides data-driven evidence for prioritizing interventions in identified hotspots and implementing integrated multi-pollutant control strategies. These insights can inform public health policy and resource allocation for air quality management.

Index Terms—Air pollution, PM2.5, Air Quality Index, Global health, Environmental data analysis

I. INTRODUCTION

A. Background

Air pollution represents one of the most pressing global health challenges of the 21st century. According to the World Health Organization (WHO), air pollution causes approximately 7 million premature deaths annually, with 91% of the world's population living in areas where air quality exceeds WHO guideline limits. The economic burden of air pollution exceeds \$5 trillion per year globally, encompassing healthcare costs, lost productivity, and environmental degradation.

Major air pollutants include particulate matter (PM2.5), carbon monoxide (CO), ozone (O₃), and nitrogen dioxide (NO₂).

B. Motivation and Research Questions

This study provides a global overview across 175 countries, identifies hotspots, analyzes pollutant relationships, and offers policy-relevant recommendations. Key questions: global AQI distribution, most impacted regions, dominant pollutants, inter-pollutant correlations, and multi-pollutant patterns.

II. METHODOLOGY

A. Dataset

Source: Global Air Pollution Dataset (see References).
Size: 23,463 records (cities).
Geographic Coverage: 175 countries.

Variables: Country, City, Overall AQI, PM2.5 AQI, CO AQI, O₃ AQI, NO₂ AQI, AQI category, measurement timestamp.

AQI categories: Good (0–50), Moderate (51–100), Unhealthy for Sensitive Groups (101–150), Unhealthy (151–200), Very Unhealthy (201–300), Hazardous (301–500).

B. Preprocessing

Data were loaded using `pandas`. Steps include missing-value and duplicate checks, range validation (0–500), standardization of column names, and flagging records at instrument ceiling (AQI=500). Any removed or corrected records are documented in the repository scripts.

C. Analysis Methods

Descriptive statistics (mean, median, std), percentile-based risk counts, country-level aggregation, and Pearson correlation coefficients between pollutants (with p-values computed). Visualizations include pie charts, bar charts, boxplots, and heatmaps. All visualizations and code are in the repository.

D. Tools

Python 3.12, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly. Work performed in Jupyter Notebook / Google Colab.

III. RESULTS

A. Global AQI Category Distribution

Table I shows the AQI category distribution. 18.4% of cities are unhealthy or worse.

TABLE I: Global AQI Category Distribution (23,463 cities)

Category	Count	Percentage
Good	9,936	42.3%
Moderate	9,231	39.3%
Unhealthy for Sensitive	1,591	6.8%
Unhealthy	2,227	9.5%
Very Unhealthy	287	1.2%
Hazardous	191	0.8%

B. Geographic Patterns

Most polluted countries (by mean AQI) and most polluted cities are shown in Figures 2 and 3.

C. Pollutant-Specific Analysis

Table II summarizes pollutant statistics.

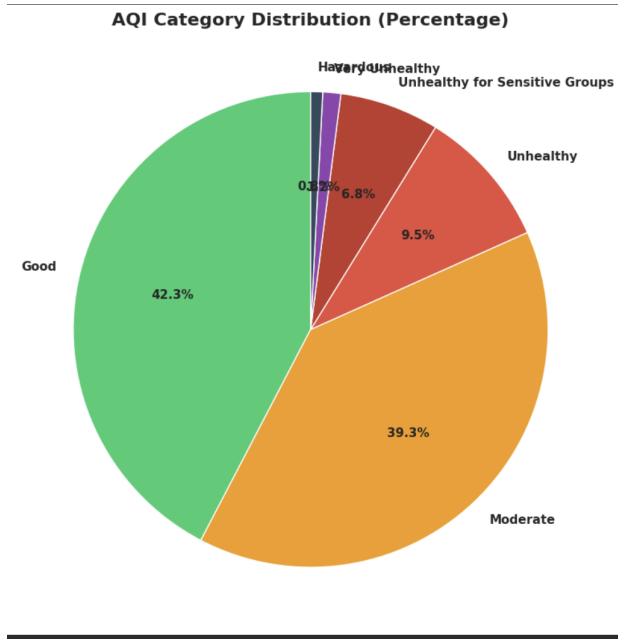


Fig. 1: Global distribution of Air Quality Index categories across 23,463 cities. Nearly one in five cities has concerning air quality levels.

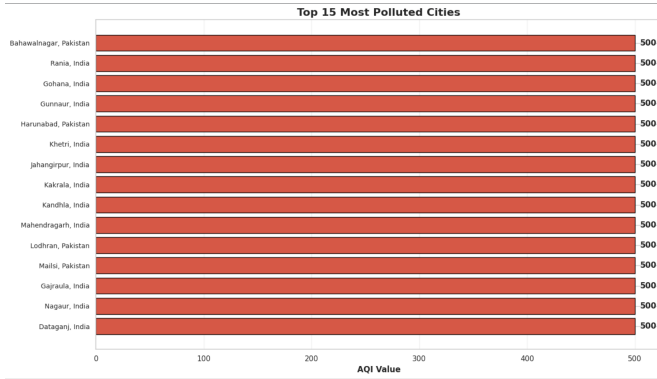


Fig. 2: Top 15 most polluted cities (horizontal bar chart). Many top cities are in India and Pakistan and some reach AQI=500.

D. PM2.5 Health Risk Assessment

PM2.5 category distribution: Good (10,208; 43.5%), Moderate (9,075; 38.7%), Unhealthy for Sensitive (1,624; 6.9%), Unhealthy (2,129; 9.1%), Very Unhealthy (255; 1.1%), Hazardous (172; 0.7%). Critical finding: 2,556 cities (10.9%) have concerning PM2.5 levels (unhealthy or worse).

E. Pollutant Correlations

Pearson correlation matrix (Table III) and heatmap (Figure 4) show moderate positive relationships between several pollutants.

F. Monitoring Coverage

Top 5 countries by number of monitored cities: United States (2,872), India (2,488), Brazil (1,562), Germany (1,345),

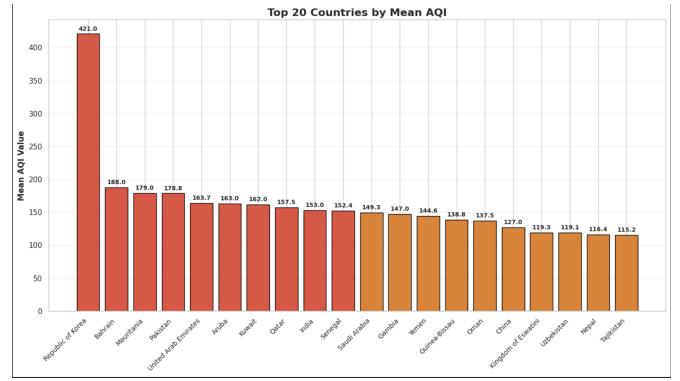


Fig. 3: Country-level mean AQI (top 20 countries by mean).

TABLE II: Pollutant Summary Statistics (AQI)

Pollutant	Mean	Median	Max	Std Dev
PM2.5	68.52	51.0	500	54.80
Ozone (O ₃)	35.19	30.0	235	28.10
NO ₂	3.06	1.0	91	5.25
CO	1.37	1.0	133	1.83

Russian Federation (1,241). Monitoring coverage shows bias toward developed countries and larger developing countries.

IV. DISCUSSION

A. PM2.5 as the Primary Threat

Our analysis identifies PM2.5 as the dominant air quality concern with broad geographic impacts concentrated in South Asia. Drivers include population density, industrialization, agricultural burning, vehicle emissions, and local meteorology.

B. Pollutant Correlations and Implications

Moderate CO–NO₂ ($r = 0.488$) and PM2.5–CO ($r = 0.439$) correlations indicate common combustion sources and point to integrated control strategies (transport, industry) as effective leverage points.

C. Ozone's Distinct Pattern

Ozone shows weaker correlations with primary pollutants due to its secondary photochemical formation, requiring precursor-focused mitigation strategies.

D. Ceiling Effect and Monitoring Gaps

Several cities reached AQI = 500 (instrument ceiling), suggesting measurements may underestimate true severity. Geographic monitoring gaps, notably in parts of Africa and South America, may mask local pollution issues.

E. Limitations

This study uses a temporal snapshot; it does not analyze seasonality. Monitoring bias and the AQI ceiling effect limit some inferences. Correlational results do not imply causation. Socioeconomic covariates were not included.

TABLE III: Pollutant Correlation Matrix (Pearson r)

	CO	Ozone	NO ₂	PM2.5
CO	1.000	0.146	0.488	0.439
Ozone	0.146	1.000	0.146	0.340
NO ₂	0.488	0.146	1.000	0.260
PM2.5	0.439	0.340	0.260	1.000

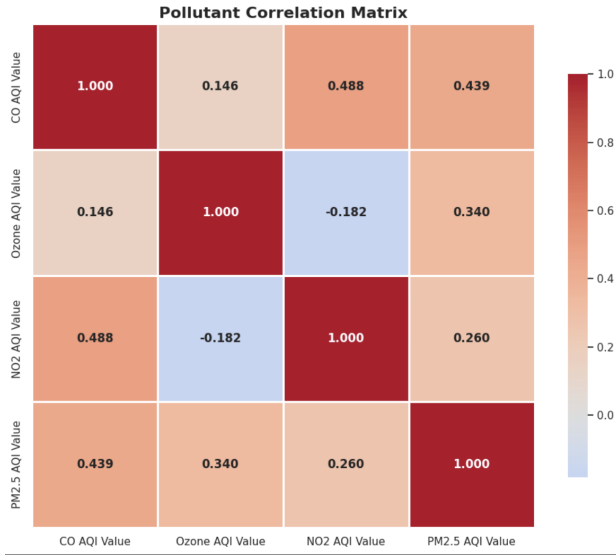


Fig. 4: Pearson correlation heatmap between pollutants.

V. CONCLUSION

This global analysis across 23,463 cities highlights PM2.5 as the primary pollutant of concern, identifies South Asia as a critical hotspot, documents moderate pollutant correlations suggesting common sources, and recommends both immediate emergency and long-term integrated policy responses. Future work should include time-series forecasting, source attribution, and health impact quantification.

ACKNOWLEDGMENTS

We thank our instructors and data providers for guidance and open data access. We also acknowledge the open-source software community used for analysis.

REFERENCES

APPENDIX

All analysis code, notebooks, and scripts are available at:

<https://github.com/divya2212001/Global-Air-Pollution>

TABLE IV: Data dictionary (sample)

Variable	Description / units
country	Country name (string)
city	City name (string)
$aqi_{overall}$	Overall AQI (0–500)
aqi_{pm25}	PM2.5-based AQI (0–500)
aqi_{co}	CO-based AQI (0–500)
aqi_{o3}	Ozone-based AQI (0–500)
aqi_{no2}	NO2-based AQI (0–500)
$measurement_{time}$	Timestamp of measurement (ISO 8601)